

Extra Hidden States under Emission Misspecification in Hidden Markov Models

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Abstract

Additional hidden Markov states can compensate for restrictive emission distributions without representing additional scientific regimes. We study this distinction using the established representation of mixture components as refined states. Under compactness, mixture identifiability and positive aggregate transitions, we prove that a simple-emission HMM with too few components has an observed-law Kullback–Leibler loss that grows at least linearly with sequence length. An exact refinement, or an aggregate HMM with mixture emissions, removes this approximation gap. Simulations illustrate the resulting incentive to add states. An analysis of tumor allele fractions on chromosome 1 then compares five Gaussian states with three regional profiles containing five Gaussian components. The fitted emissions and regional assignments are nearly the same after grouping the Gaussian states, while the mixture model uses fourteen fewer parameters. This example illustrates how additional states can describe variation within a regional profile rather than additional regional regimes. The number of fitted states alone therefore does not determine the number of scientifically distinct regimes.

Keywords: hidden Markov model; emission misspecification; finite mixture; relative-entropy rate; state splitting; allelic imbalance.

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1 Introduction

Hidden Markov models (HMMs) separate serial dependence from the conditional distribution of observations through a latent state process. This decomposition supports applications across time-series analysis, signal processing, and ecology (Rabiner, 1989; Cappé et al., 2005; Zucchini et al., 2016). Fitted states are often labelled as behaviours, regimes, phases, or classes and then become the main scientific object reported from an analysis (McClintock et al., 2020).

That interpretation is useful but not automatic. A fitted state is also part of a statistical approximation. If the state-dependent distribution is too restrictive, likelihood can improve by

creating states whose emissions jointly approximate skewness, heavy tails, overdispersion, or within-regime heterogeneity. An additional state can therefore be statistically useful without representing an additional scientific regime. Under misspecification, likelihood targets a best approximation in the fitted family rather than the data-generating parameter (White, 1982; Douc and Moulines, 2012).

The practical phenomenon is established in the HMM literature. Pohle et al. (2017) show that information criteria can favour state counts that are difficult to interpret scientifically. Additional states may absorb unmodelled structure in state-dependent distributions, while flexible parametric, mixture, and nonparametric observation models offer alternatives (Dannemann, 2012; Dannemann et al., 2014; Langrock et al., 2015, 2018). The distinction matters because methods for estimating statistical HMM order (Gassiat and Keribin, 2000; Gassiat and Rousseau, 2014; Lehericy, 2019; Zou et al., 2024; Lin and Huang, 2025) do not imply that every selected state is a distinct scientific regime.

Mixture-emission estimation and component aggregation already have a substantial methodological basis. Volant et al. (2014) give the component-state representation and an EM-based merging approach. Holzmänn and Schwaiger (2015) characterize minimal representations and test whether states can be reduced to mixture components without changing the observed law. We use this established refinement construction as the starting point, not as a new model or estimation algorithm.

Our result concerns uniform separation from a restrictive emission class. Under compactness, identifiable mixture order and positive aggregate transitions, every one-step true predictive density remains a positive L^1 distance from every mixture with only the aggregate number of simple components. The chain rule turns this bound into a Kullback–Leibler divergence that is at least proportional to sequence length against every HMM in the smaller class, without aligning hidden paths or restricting candidate transitions. Marginal mixture identifiability already establishes non-representability. The additional predictive bound gives uniform separation at every sequence length and a positive relative-entropy-rate gap. The result complements HMM likelihood and identifiability theory (Leroux, 1992; Bickel et al., 1998; Allman et al., 2009; Gassiat et al., 2016; Alexandrovich et al., 2016).

The empirical comparison asks whether an additional state is needed to describe a new regime or to represent variation within an existing one. We compare ordinary and mixture-emission HMMs, including pairs with the same total number of Gaussian components. Holding this count fixed isolates the additional freedom in the transition model. A simulation shows the mechanism when the aggregate states are known. The application examines chromosome-1 tumor allele fractions: low and high fractions at successive markers can describe the same regional imbalance. This gives a scientific reason to group two emission components within one Markov state. The Supporting Information contains further genomic comparisons and an elk movement example that examines sensitivity to the emission family.

Section 2 gives the formal mechanism; Sections 3 and 4 describe estimation and model comparison;

and Sections 5 and 6 illustrate the comparison in simulation and data.

2 Why restrictive emissions create extra states

If a scientifically meaningful regime has a heterogeneous emission distribution, several simple-emission states can improve fit by representing its components rather than by revealing new regimes. The formal results below give conditions under which an aggregate-state model with richer emissions and a refined model with additional simple-emission states generate the same observed process.

Let $\mathbf{Y}$ be the observation set, let $\mathcal{B}$ be its sigma-field, let λ be a dominating measure on $(\mathbf{Y}, \mathcal{B})$, and let Θ be the emission-parameter space. All densities below are with respect to λ . For densities p and q , define the Kullback–Leibler divergence and the $L^1(\lambda)$ distance by

$$D(p\|q) = \int_{\mathbf{Y}} p(y) \log \left\{ \frac{p(y)}{q(y)} \right\} \lambda(dy), \quad \|p - q\|_1 = \int_{\mathbf{Y}} |p(y) - q(y)| \lambda(dy).$$

Here, $0 \log(0/q) = 0$, and $D(p\|q) = +\infty$ if $q = 0$ on a set having positive probability under p . The same notation $D(P\|Q)$ is used for probability laws through their densities. Let $\mathcal{F} = \{f_\eta : \eta \in \Theta\}$ be the simple fitted emission family. Define the class of mixtures with at most K atoms, for an integer $K \geq 1$, by

$$\mathcal{M}_K = \left\{ \sum_{j=1}^K \omega_j f_{\eta_j} : \omega_j \geq 0, \sum_{j=1}^K \omega_j = 1, \eta_j \in \Theta \right\}.$$

Thus, the ω_j are mixing proportions. Zero weights are permitted, so $\mathcal{M}_K \subseteq \mathcal{M}_{K+1}$. Let $\mathbf{H}_K(\mathcal{F})$ denote the observed process laws generated by K -state HMMs with emissions in $\mathcal{F}$, arbitrary transition matrices, and arbitrary initial distributions. For process laws P and Q , let $P^{1:T}$ and $Q^{1:T}$ denote the respective joint laws of $(Y_1, \dots, Y_T)$ and write

$$\underline{D}^{\text{rate}}(P\|Q) = \liminf_{T \rightarrow \infty} T^{-1} D(P^{1:T}\|Q^{1:T})$$

for their lower relative-entropy rate.

Fix an integer $K_0 \geq 1$, interpreted as the number of aggregate scientific regimes. A superscript 0 denotes a data-generating parameter. The following assumptions isolate a transparent finite-mixture setting.

Assumption 2.1 (Emission regularity). The parameter space Θ is compact, the map $\eta \mapsto f_\eta$ is continuous into $L^1(\lambda)$, and every f_η is positive λ -almost everywhere.

Assumption 2.2 (Identifiable mixture order). Finite mixtures from $\mathcal{F}$ have identifiable minimal order. Two equal finite mixtures with positive weights and pairwise distinct parameters have the same number of components and agree up to permutation.

Classical sufficient conditions are given by [Teicher \(1963\)](#) and [Yakowitz and Spragins \(1968\)](#); see also [McLachlan and Peel \(2000\)](#). The assumptions include univariate Gaussian emissions on compact mean and variance sets with variances bounded away from zero.

Lemma 2.3 (Separation from lower-order mixtures). *Under Assumptions 2.1 and 2.2, let $R > K_0$ be an integer and let $\eta_1^0, \dots, \eta_R^0 \in \Theta$ be pairwise distinct. For $c > 0$ and $Rc \leq 1$, set*

$$\mathcal{A}_c = \left\{ \sum_{r=1}^R \omega_r f_{\eta_r^0} : \omega_r \geq 0, \sum_{r=1}^R \omega_r = 1 \right\}.$$

Then $d_c := \inf_{p \in \mathcal{A}_c} \inf_{q \in \mathcal{M}_{K_0}} \|p - q\|_1 > 0$, and

$$\inf_{p \in \mathcal{A}_c} \inf_{q \in \mathcal{M}_{K_0}} D(p \| q) \geq d_c^2 / 2 > 0.$$

The lemma says that a mixture containing R distinct atoms, each with a non-negligible weight, cannot be approximated arbitrarily closely by only $K_0 < R$ atoms.

To see the approximation incentive directly, let $Z_t \in \{1, \dots, K_0\}$ be a stationary aggregate Markov chain with transition matrix $\Gamma^0 = (\gamma_{k\ell}^0)$ and stationary probabilities $\pi_k = \mathbb{P}(Z_t = k) > 0$. Here, $\gamma_{k\ell}^0 = \mathbb{P}(Z_t = \ell \mid Z_{t-1} = k)$. Suppose the emission in state k is g_k , and, for an integer $m \geq 1$, define $\delta_k(m) = \inf_{q \in \mathcal{M}_m} D(g_k \| q)$. Thus, $\delta_k(m)$ is the infimum of the emission-level Kullback–Leibler loss with at most m atoms from $\mathcal{F}$. For specified aggregate-state emissions $q_1, \dots, q_{K_0}$, define the aligned risk by

$$\mathcal{R}(q_1, \dots, q_{K_0}) = \sum_{k=1}^{K_0} \pi_k D(g_k \| q_k).$$

This comparison fixes the law of the aggregate hidden trajectory. In the refined representation, the within-state component is drawn independently at each visit, conditional on the aggregate states; the emission q_k can therefore be any member of $\mathcal{M}_{m_k}$.

Proposition 2.4 (Aligned improvement by state splitting). *If aggregate state k is represented by one emission from $\mathcal{F}$, the infimum of the aligned Kullback–Leibler risk is $\sum_{k=1}^{K_0} \pi_k \delta_k(1)$. If it is represented by $m_k \geq 1$ substates, where m_k is the number allocated to aggregate state k , the infimum of the aligned risk is $\sum_{k=1}^{K_0} \pi_k \delta_k(m_k)$. If $\sum_k \pi_k \delta_k(1) < \infty$, the second risk is strictly smaller whenever $\delta_{k^*}(m_{k^*}) < \delta_{k^*}(1)$ for at least one state.*

State splitting gives a heterogeneous state several emission atoms instead of one. The statement is only an aligned comparison because it holds the aggregate trajectory law fixed; attainment of the infima is not required. The next construction removes that restriction.

We recall the component-state construction of Volant et al. (2014); see also Holzmänn and Schwaiger (2015). Suppose each aggregate emission is a finite mixture with m_k components and $\eta_{k,a}^0 \in \Theta$,

$$g_k(y) = \sum_{a=1}^{m_k} \omega_{k,a}^0 f_{\eta_{k,a}^0}(y), \quad \omega_{k,a}^0 > 0, \quad \sum_a \omega_{k,a}^0 = 1. \quad (1)$$

The $\omega_{k,a}^0$ are the within-state mixing proportions.

Proposition 2.5 (Exact HMM refinement). *Let the refined states be (k, a) , where $1 \leq k \leq K_0$ and*

$1 \leq a \leq m_k$. Give state (k, a) emission $f_{\eta_{k,a}^0}$ and transition probabilities

$$\Gamma_{(k,a),(\ell,b)}^{\text{ref}} = \gamma_{k\ell}^0 \omega_{\ell,b}^0. \quad (2)$$

With initial probabilities $\pi_k \omega_{k,a}^0$, the refined HMM has exactly the same observed process law as the aggregate HMM with transitions Γ^0 and emissions in (1).

Equation (2) retains the probability of moving from aggregate state k to ℓ , then draws a component of the destination emission. The refined states therefore encode within-regime mixture components, not new aggregate dynamics.

For the genomic example in Section 6, the component counts $(2, 2, 1)$ give three aggregate states and five refined states. The two components of each paired state have different emissions but share the same transition row in the refined representation. An unrestricted five-state HMM can instead give each component its own transition row. This is the specific restriction examined in the application.

Assume now that $\varepsilon := \min_{1 \leq k, \ell \leq K_0} \gamma_{k\ell}^0 > 0$, let $R = \sum_{k=1}^{K_0} m_k$, let $\omega_{\min} = \min_{1 \leq k \leq K_0, 1 \leq a \leq m_k} \omega_{k,a}^0$, and set $c = \varepsilon \omega_{\min}$. Suppose $R > K_0$ and all $\eta_{k,a}^0 \in \Theta$ are pairwise distinct.

Theorem 2.6 (Observed-law separation). *Under Assumptions 2.1 and 2.2, the stationary aggregate HMM with emissions (1) has observed law P_0^Y . For every integer $T \geq 1$ and every $Q \in \mathbf{H}_{K_0}(\mathcal{F})$,*

$$D\{(P_0^Y)^{1:T} \| Q^{1:T}\} \geq T d_c^2 / 2. \quad (3)$$

Consequently,

$$\inf_{Q \in \mathbf{H}_R(\mathcal{F})} \underline{D}^{\text{rate}}(P_0^Y \| Q) = 0, \quad \inf_{Q \in \mathbf{H}_{K_0}(\mathcal{F})} \underline{D}^{\text{rate}}(P_0^Y \| Q) \geq d_c^2 / 2 > 0.$$

The constant is uniform over the candidate HMMs, but depends on the fixed true atoms, their minimum weight, the transition bound and the compact fitted parameter set. It need not stay bounded away from zero when distinct atoms approach one another or these lower bounds vanish.

The first equality follows because the refined R -state model reproduces the truth exactly. For the second, every one-step prediction from a K_0 -state simple-emission HMM belongs to $\mathcal{M}_{K_0}$, while every true prediction contains all R atoms with weights at least c . The mixture separation lemma and the chain rule for relative entropy give (3) for every T , hence the positive rate gap. This argument concerns observed predictive distributions and does not align hidden paths.

More generally, for any integer $1 \leq K < R$, define $d_{c,K} := \inf_{p \in \mathcal{A}_c} \inf_{q \in \mathcal{M}_K} \|p - q\|_1 > 0$. The same argument gives $D\{(P_0^Y)^{1:T} \| Q^{1:T}\} \geq T d_{c,K}^2 / 2$ for every $Q \in \mathbf{H}_K(\mathcal{F})$. The constant depends on K ; $d_{c,K_0} = d_c$. Thus the approximation gap persists whenever the candidate has fewer states than the true number of distinct emission components, not only when it has K_0 states. Supporting Information S1.4 gives the argument for this extension.

Corollary 2.7 (Enriched-emission representation). *If the fitted emission family is enlarged from $\mathcal{F}$ to $\mathcal{M}_M$, with an integer $M \geq \max_{1 \leq k \leq K_0} m_k$, then*

$$\inf_{Q \in \mathbf{H}_{K_0}(\mathcal{M}_M)} \underline{D}^{\text{rate}}(P_0^Y \| Q) = 0.$$

Thus, a refined model with more simple-emission states and an aggregate model with richer emissions can describe the same observations. Equality of the two representations does not determine which latent description is scientifically meaningful.

A likelihood-selection consequence requires an additional stochastic limit, not just a bound on expected divergence. For example, suppose the normalized log-likelihood ratios $T^{-1} \log\{p_0(Y_{1:T})/q(Y_{1:T})\}$ converge almost surely, uniformly over $Q \in \mathbf{H}_{K_0}(\mathcal{F})$, to limits bounded below by $d_c^2/2$. Then the maximized log-likelihood advantage of the R -state class is at least $T(d_c^2/2 - o(1))$, since this class contains the truth. Any fixed parameter-count BIC penalty difference is only of order $\log T$. Under that additional assumption, BIC eventually prefers R to K_0 simple-emission states. This conditional comparison neither proves the uniform convergence assumption nor selects R against every other order.

For reference, Sections S1.1–S1.5 of the Supporting Information give proofs of Lemma 2.3, Proposition 2.4, Proposition 2.5, Theorem 2.6, and Corollary 2.7, respectively.

3 Fitting the enriched-emission HMM

Let $X_t \in \{1, \dots, K\}$ denote the fitted hidden state at time t . For state k , let $M_k \geq 1$ be the number of fitted Gaussian components and let $\phi(y; \mu, \sigma^2)$ denote the Gaussian density at y with mean μ and variance $\sigma^2 > 0$. The fitted state-dependent density is

$$b_k(y) = \sum_{m=1}^{M_k} \omega_{km} \phi(y; \mu_{km}, \sigma_{km}^2), \quad \omega_{km} \geq 0, \quad \sum_m \omega_{km} = 1.$$

The quantities ω_{km} are mixing proportions within state k ; they do not change the state transition probabilities. The hidden chain has initial probabilities $\alpha_k = \mathbb{P}(X_1 = k)$ and transition probabilities $a_{k\ell} = \mathbb{P}(X_t = \ell \mid X_{t-1} = k)$, collected in $A = (a_{k\ell})$.

The following standard mixture-emission expectation-maximization (EM) updates are included to specify the fitted model (Volant et al., 2014). The algorithm alternates between state and component allocation and closed-form parameter updates. Write $Y_{1:T} = (Y_1, \dots, Y_T)$ and let $C_t \in \{1, \dots, M_{X_t}\}$ denote the Gaussian-component label at time t . A scaled forward–backward recursion gives

$$\begin{aligned} \zeta_t(k) &= \mathbb{P}(X_t = k \mid Y_{1:T}), \\ \xi_t(k, \ell) &= \mathbb{P}(X_{t-1} = k, X_t = \ell \mid Y_{1:T}). \end{aligned}$$

Here, $\zeta_t(k)$ is defined for $t = 1, \dots, T$, whereas $\xi_t(k, \ell)$ is defined for $t = 2, \dots, T$. The joint posterior

responsibility for state k and component m is

$$r_t(k, m) = \mathbb{P}(X_t = k, C_t = m \mid Y_{1:T}) = \zeta_t(k) \frac{\omega_{km} \phi(Y_t; \mu_{km}, \sigma_{km}^2)}{b_k(Y_t)}.$$

The M-step updates are

$$\begin{aligned} \alpha_k^{\text{new}} &= \zeta_1(k), & a_{k\ell}^{\text{new}} &= \frac{\sum_{t=2}^T \xi_t(k, \ell)}{\sum_{t=2}^T \sum_j \xi_t(k, j)}, \\ \omega_{km}^{\text{new}} &= \frac{\sum_t r_t(k, m)}{\sum_t \zeta_t(k)}, & \mu_{km}^{\text{new}} &= \frac{\sum_t r_t(k, m) Y_t}{\sum_t r_t(k, m)}, \\ (\sigma_{km}^2)^{\text{new}} &= \frac{\sum_t r_t(k, m) (Y_t - \mu_{km}^{\text{new}})^2}{\sum_t r_t(k, m)}. \end{aligned}$$

In these updates, sums over t without displayed limits run from 1 to T , and $\sum_j$ runs over the K fitted states. For independent sequences with common parameters, the forward-backward recursion resets at each sequence start. The initial-probability update is the average of the state responsibilities at those starts; transition sums exclude boundaries, while emission sums pool all observations. Numerical pseudocounts and variance constraints are specified in SI S2–S3. In computation, a small lower bound on σ_{km} prevents likelihood degeneracy. Multiple starts are needed because mixture-HMM likelihoods are multimodal. For a sequence of length T , BIC is $-2\ell(\hat{\theta}) + p \log T$, where $\hat{\theta}$ is the maximum likelihood estimate, $\ell(\hat{\theta})$ is the maximized log-likelihood, and the number p of free parameters is

$$(K - 1) + K(K - 1) + \sum_{k=1}^K \{(M_k - 1) + 2M_k\}.$$

The three terms count initial probabilities, transition probabilities, and state-specific mixture parameters, respectively.

4 Comparing emission and transition complexity

Suppose scientific knowledge suggests K_0 aggregate regimes, while a simple-emission analysis favours additional states. A K_0 -state HMM with $R = \sum_k M_k$ Gaussian components is an R -state Gaussian HMM whose transition probabilities satisfy the refinement constraints in Proposition 2.5, with initial probabilities $\alpha_{(k,m)}^{\text{ref}} = \alpha_k \omega_{km}$. Comparing it with an unrestricted R -state model holds the emission component count fixed and changes the latent dynamics. For example, two aggregate states with component counts $(2, 1)$ have 10 free parameters, compared with 14 for three Gaussian states; counts $(2, 2)$ give 13, compared with 23 for four Gaussian states. A comparison of two states with four components against three Gaussian states changes both the component count and the transition structure.

The factorization requires both equal transition rows within each source group and destination-

component proportions that do not depend on the source state. Equal rows or a low matrix rank alone do not establish the full constraint (Holzmann and Schwaiger, 2015). We specify the groups from the emission mechanism, then compare the constrained and unrestricted fits. The state-reduction test of Holzmann and Schwaiger (2015) addresses this question under a stationary initial law. For the genomic comparison we use a model-conditional parametric bootstrap with the fitted initial law; Supporting Information S3.2 explains its calibration and limitations.

BIC summarizes penalized likelihood within the specified model classes. It is not a test of the number of biological regimes. In particular, mixture singularities and transition estimates near the boundary limit a regular asymptotic interpretation of its penalty. An unrestricted refined fit must have likelihood at least as high as the corresponding constrained fit at a common variance bound; initialization at the exact refinement provides a numerical check of that nesting.

Prediction is assessed separately from fit. Parameters estimated on the training data are held fixed while the filter incorporates observations in an evaluation sequence. Its sequential log score is $\sum_t \log p_{\hat{\theta}}(Y_t | Y_{1:t-1})$, with the fitted initial law used at a new sequence start. Higher scores indicate better predictive densities on that sequence. For a continuation of a training sequence, the filter instead continues from the training endpoint. Probability integral transforms and interval coverage provide further descriptive checks in the Supporting Information. Similar log scores alone do not establish predictive equivalence.

Decoded sequences and state-dependent distributions connect these comparisons to the observations. Cross-tabulations indicate when two Gaussian states map largely to one mixture-emission state, but do not establish that the latter is scientifically correct. States index Markov dynamics; mixture components describe within-state distributions; modes are local maxima of a density. These three counts need not coincide. Interpretation therefore also depends on temporal stability, sensitivity to fitting constraints, and measurements relevant to the scientific question.

5 Simulation study

We write GK for a K -state Gaussian HMM. For mixture-emission models, the digits after M give the component counts in successive states: $M21$ has counts $(2, 1)$, $M22$ has $(2, 2)$, and $M221$ has $(2, 2, 1)$.

We first isolate the mechanism in a controlled setting. The data-generating process has two aggregate states, transition matrix

$$\Gamma^0 = \begin{pmatrix} 0.93 & 0.07 \\ 0.05 & 0.95 \end{pmatrix}.$$

The first-state emission was a two-component Gaussian mixture with weights 0.70 and 0.30, means -1.20 and 1.10 , and standard deviations 0.55 and 0.45. The second-state emission was Gaussian with mean 3.80 and standard deviation 0.65. The process therefore has two aggregate regimes and three distinct Gaussian atoms.

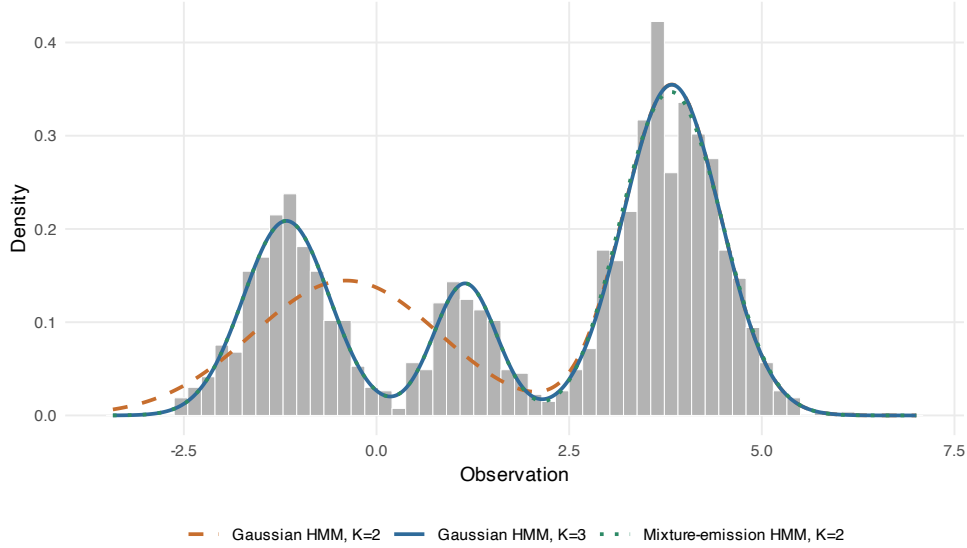


Figure 1: Observed histogram and fitted marginal densities for the reference simulation. The two-state Gaussian model smooths over the heterogeneous first regime. The three-state Gaussian and enriched two-state models recover its two components.

For a reference series of length 1500, we fitted Gaussian HMMs with $K = 2$ and $K = 3$, and an enriched $K = 2$ HMM with two Gaussian components per state. Each model was fitted with the protocol used for the replications below. The two-state Gaussian model merged the heterogeneous first regime and had BIC 4487.4. The three-state Gaussian model separated its two components and had BIC 4197.3. The enriched two-state model recovered the same within-regime structure while retaining two aggregate states and had BIC 4191.6. Figure 1 shows the directly observable emission fit.

Across 200 fixed-seed replications at each sample size, BIC preferred the three-state to the two-state Gaussian model in 99.5%, 100% and 100% of samples for $T = 500, 1500, 5000$, respectively. Comparing G3 with the enriched two-state model, BIC preferred the latter in 96.0%, 99.0% and 99.5%. Both classes contain the data-generating law. Their fitted log likelihoods are nearly equal on average, and the mean M22-minus-G3 BIC differences $-6.5, -7.7, -8.6$ are close to $-\log T$. The preference mainly reflects M22’s smaller parameter count (13 versus 14), rather than a substantial improvement in fit. An initial fitting protocol preferred G2 over G3 by BIC in 40 of the 600 replications. The reported fits use additional data-derived starts and refinement to relative likelihood tolerance 10^{-9} . In particular, an exact refinement of a fitted M21 model supplies a G3 start without using generating parameters. This improved optimization changes 39 of those 40 preferences to G3: the initial G2 preferences mainly reflected incomplete optimization. SI S2 documents both protocols and the retained fits. The enriched candidate has four components although the data-generating distribution has three; these frequencies therefore do not isolate the equal-component comparison of Section 4. That comparison is implemented directly in the genomic analysis. Full computational details and additional diagnostics are reported in the Supporting Information.

These frequencies are descriptive. Finite mixtures and overfitted HMMs contain singular parameter strata, so regular-model interpretations of BIC require caution (Keribin, 2000; Rousseau and Mengersen, 2011; Gassiat and Rousseau, 2014; Watanabe, 2013; Drton and Plummer, 2017).

6 Chromosomal allelic imbalance

6.1 Question and models

We analyze chromosome 1 of the breast-cancer cell line HCC1143 and its matched normal cell line HCC1143BL, using the SNP-array measurements of Chiang et al. (2009). At a single-nucleotide polymorphism (SNP), the B-allele fraction is the B signal divided by the combined A and B signals. The measurements are unphased: the allele labelled B need not come from the same parental chromosome at successive markers. Consequently, fractions near 0.2 and 0.8 can describe the same regional allelic imbalance (Olshen et al., 2011). A mixture emission can represent these two bands within one regional profile. The question is whether the bands require separate transition dynamics. Data details are given in Supporting Information S3.1.

We fit G2–G5, M21 and M221 to 20,697 ordered markers. Each Markov state of a mixture model represents a regional profile, corresponding to an aggregate regime in Section 2. The M21/G3 and M221/G5 comparisons hold the total number of Gaussian components fixed. In particular, M221 corresponds to $K_0 = 3$ and $R = 5$ in Proposition 2.5. G5 relaxes its component-group transition and initial-distribution constraints, adding 14 parameters without adding emission components. If an M221 process satisfied the assumptions of Theorem 2.6, its extension to $K < R$ would give a positive approximation gap for G2–G4. This motivates examining emission complexity; the fitted data do not verify those assumptions. Multiple starts and numerical checks are documented in Supporting Information S3.2.

6.2 Chromosome-1 results

Table 1: Chromosome-1 comparison. K is the number of Markov states, R the total number of Gaussian components, and p the number of free parameters. Lower BIC is preferred. Repeat scores are sequential log scores on chromosome 1 of the second array pair, with fitted parameters held fixed; higher scores are preferred. This is a technical repeat of the same cell lines.

Model	K	R	p	BIC	Repeat score
G2	2	2	7	-14751.5	7049.59
G3	3	3	14	-20738.2	9818.62
G4	4	4	23	-22415.1	10767.74
G5	5	5	34	-24697.0	11907.57
M21	2	3	10	-20773.1	9816.15
M221	3	5	20	-24817.8	11906.57

M221 estimates paired component centres at (0.185, 0.805) and (0.392, 0.601), and a single centre at 0.498. We call these strong imbalance, moderate imbalance and approximate balance, respectively.

These names describe the measured fractions, not cell populations or absolute DNA copy numbers. Figure 2 shows both the regional profiles and the Gaussian components within them.

G4 improves on G3, but its BIC remains 2402.7 higher than M221's. Among G2–G5, G5 has the lowest BIC. The more informative comparison for the refinement construction is G5 versus M221: G5 estimates almost the same five component distributions, gains only 9.13 log-likelihood units, and has a BIC 120.87 higher. Thus, on this chromosome the additional component-level transition freedom produces little improvement in fit. A model-conditional parametric bootstrap under M221 gives a likelihood-ratio statistic of 18.26 and a Monte Carlo p-value of 0.09 from 199 replicates (Supporting Information S3.2), without rejecting the refinement constraints at the 5% level.

The raw G5 Viterbi path has 9,123 runs, compared with 181 for M221. Many G5 switches are between the low and high bands within the same regional profile. Matching the Gaussian emissions and grouping the paired G5 states gives 99.73% agreement with M221 and 194 runs. The ordinary HMM therefore recovers nearly the same regional description after aggregation; the mixture model represents that grouping directly in its Markov structure.

A second array pair measures the same cell lines again. With the chromosome-1 estimates held fixed, M221 and G5 have almost identical sequential log scores on this technical repeat (Table 1). The M221 paths decoded from the first and second array pairs assign the same regional profile to 98.08% of the 19,727 shared markers. These results assess repeatability within this cell line, not performance across patients or accuracy against known regional states.

The application illustrates the distinction between emission components and regional dynamics. It does not establish that three is the true number of regional regimes. Unequal marker spacing and residual dependence also limit the interpretation of individual short runs. Supplementary analyses include prediction and calibration (S3.3), repeatability and sensitivity (S3.4), models incorporating allelic symmetry (S3.5), and comparisons with genomic segmentation and independent sequencing (S3.6). S3.3 identifies evaluation sequences on which other models predict better; S3.5 reports the fit and prediction trade-offs of imposing symmetry. S3.6 examines how the inferred profiles relate to measurements from other methods.

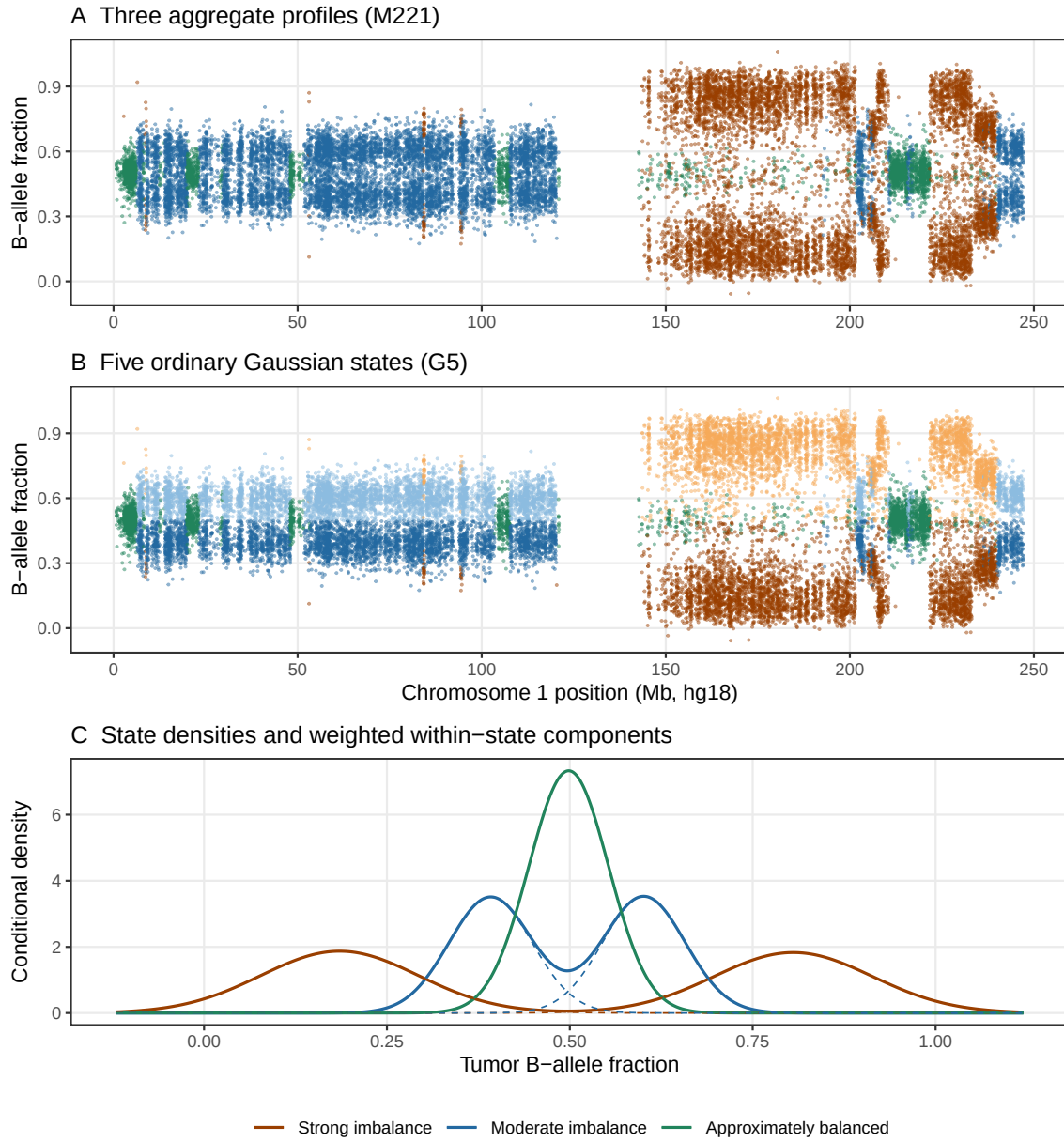


Figure 2: Chromosome 1 of HCC1143. Panels A and B colour allele fractions by the M221 and G5 Viterbi paths. Paired light and dark colours in B distinguish Gaussian states belonging to the same aggregate colour in A. Panel C shows M221 state densities (solid) and weighted component contributions (dashed). The densities are conditional on the aggregate state.

7 Discussion

The theoretical result separates the number of emission components from the number of aggregate regimes. When the aggregate emissions contain more identifiable components than an ordinary HMM can supply, the assumptions of Theorem 2.6 imply a positive predictive approximation gap. Extra simple-emission states can remove this gap through the exact refinement construction. They need not introduce new aggregate dynamics.

The simulation illustrates this mechanism with known aggregate states. The genomic application examines its practical counterpart: whether low and high allele-fraction bands need separate regional transition laws. M221 and G5 fit nearly the same component distributions and yield nearly the same regional assignments after aggregation. M221 makes the grouping explicit and uses fewer parameters. The data illustrate the construction; they do not verify all assumptions of the separation theorem or identify a unique biological state count.

Comparisons should therefore distinguish improved emission fit from evidence for additional regimes. At a fixed component count, the constrained and unrestricted models isolate additional freedom in the latent dynamics. BIC, predictive scores and decoded paths then answer different questions. On chromosome 1, M221 has the lower BIC while G5 has a slightly higher technical-repeat score (Table 1); neither criterion alone establishes the regional interpretation. In particular, nonrejection of the refinement constraints in the model-conditional bootstrap is not an equivalence result.

The formal result assumes finite mixtures, a compact fitted emission family and positive aggregate transitions. Its separation constant can shrink as components approach one another or lower weight and transition bounds vanish. Heavy-tailed or skewed emissions may require increasingly rich approximations rather than a finite exact refinement. Structural zeros would require an accessibility argument in place of the uniform one-step bound. The theorem also does not establish finite-sample state recovery or BIC consistency; the likelihood-selection implication stated in Section 2 requires an additional uniform stochastic limit.

More generally, extra states can absorb omitted covariates, misspecified transitions or unresolved temporal structure as well as restrictive emissions. The supplementary elk analyses illustrate sensitivity to these modeling choices. Scientific interpretation therefore requires domain knowledge and measurements beyond fitted state labels (Pohle et al., 2017; McClintock et al., 2020). Neither equivalent observed laws nor similar fitted partitions determine a unique scientific interpretation of the hidden process.

8 Conclusion

A simple-emission HMM can gain useful states because its emission family is too restrictive. Exact refinement and the predictive separation bound explain how this can occur without additional aggregate regimes. Comparing ordinary and mixture-emission models at a fixed component count makes the extra freedom in latent dynamics explicit. In the chromosome-1 example, M221 represents

three regional profiles using five Gaussian components. Grouping the five G5 states gives nearly the same regional assignments. The scientific meaning of those profiles must be assessed separately from the number of fitted states.

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Data availability statement

The SNP-array data are publicly available in Gene Expression Omnibus, accession GSE13372, at <http://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE13372>. Sequencing segment estimates are in the supplementary materials of Chiang et al. (2009). The accompanying reproducibility package contains processed marker tables, fitted models, scripts, fixed seeds, source hashes and numerical tests for the simulations, genomic application and supplementary elk analyses. It documents regeneration from the original CEL files and annotation files, as well as reproduction from the supplied processed inputs. The project repository is <https://github.com/AurelienNicotiaULaval/corrective-hmm-states>. The package documents the original sources for each data set.

Ethics statement

This study reanalyzes publicly available cell-line measurements and previously published animal data. No new participants were recruited, no new biological samples were collected and no animals were handled for this study.

Conflict of interest statement

The author declares no conflict of interest.

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Figure legends

Figure 1. Observed histogram and fitted marginal densities for the reference simulation. The two-state Gaussian model smooths over the heterogeneous first regime. The three-state Gaussian and enriched two-state models recover its two components.

Figure 2. Chromosome 1 of HCC1143. Panels A and B colour allele fractions by the M221 and G5 Viterbi paths. Paired light and dark colours in B distinguish Gaussian states belonging to the same aggregate colour in A. Panel C shows M221 state densities (solid) and weighted component contributions (dashed). The densities are conditional on the aggregate state.